

CROWD MANAGEMENT SYSTEM WITH SAFE ZONE RECOMMENDATION FOR CHILDREN AND ELDERLY

A PROJECT REPORT

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ABSTRACT

Large-scale public gatherings at places such as temples, railway stations, stadiums, and festivals often lead to overcrowding and safety risks. Traditional monitoring systems rely on manual CCTV observation, which is slow, errorprone, and ineffective during emergencies. Additionally, most existing automated systems focus only on crowd density and fail to consider vulnerable groups like children and the elderly. Therefore, an intelligent and real-time crowd monitoring solution is essential.

This project proposes system that uses deep learning and computer vision techniques to analyze live CCTV video feeds. The system divides each scene into zones and evaluates them individually using models such as CSRNet for density estimation, YOLOv8 for human detection, and OpenCV for classifying individuals based on age groups. Higher importance is given to vulnerable groups to improve safety assessment. Each zone is assigned a risk score by combining crowd density and vulnerability factors, and zones are categorized as safe, moderate, or high risk.

The system also includes a graph-based routing mechanism using the A* algorithm to identify the safest evacuation paths from risky areas to exits. A Streamlit dashboard provides real-time visualization, alerts, and guidance for authorities. In critical situations, an emergency alert mode is activated.

Overall, the proposed system offers an efficient and scalable solution for crowd monitoring by integrating detection, risk analysis, and safe route recommendation, thereby improving public safety in large gatherings.

திட்டப்பணிச்சுருக்கம்

கோவில்கள், ரயில் நிலையங்கள், விளையாட்டு அரங்குகள் போன்ற இடங்களில் மக்கள் அதிகமாக கூடுவதால் பாதுகாப்பு சிக்கல்கள் உருவாகின்றன. தற்போதைய முறைகள் மனிதர்களின் CCTV கண்காணிப்பை மட்டுமே சார்ந்துள்ளதால் அவை தாமதமாகவும் குறைந்த திறனுடனும் செயல்படுகின்றன. மேலும், குழந்தைகள் மற்றும் முதியவர்கள் போன்ற பலவீனமான குழுக்களுக்கு தனியான கவனம் வழங்கப்படுவதில்லை. எனவே, நிகழ்நேரத்தில் செயல்படும் நுண்ணறிவு கூட்ட நிர்வாக அமைப்பு தேவையாகிறது.

இந்த திட்டம் CrowdGuard எனப்படும் பார்வை அடிப்படையிலான கூட்ட நிர்வாக அமைப்பை முன்வைக்கிறது. இது பல CCTV வீடியோ தரவுகளை பயன்படுத்தி, காட்சிகளை மண்டலங்களாகப் பிரித்து தனித்தனியாக பகுப்பாய்வு செய்கிறது. CSNet மூலம் கூட்ட அடர்த்தி கணக்கிடப்படுகிறது, YOLOv8 மூலம் நபர்கள் கண்டறியப்படுகின்றனர், மற்றும் OpenCV மூலம் வயது அடிப்படையில் வகைப்படுத்தப்படுகின்றனர். குழந்தைகள் மற்றும் முதியவர்களுக்கு அதிக முக்கியத்துவம் வழங்கப்படுகிறது.

ஒவ்வொரு மண்டலத்திற்கும் ஆபத்து மதிப்பெண் கணக்கிடப்பட்டு, அது பாதுகாப்பானது, மிதமானது அல்லது ஆபத்தானது என வகைப்படுத்தப்படுகிறது. A* algorithm பயன்படுத்தி பாதுகாப்பான வெளியேற்ற வழிகள் கண்டறியப்படுகின்றன. Streamlit டாஷ்போர்டு மூலம் நேரடி காட்சி, எச்சரிக்கை மற்றும் வழிகாட்டுதல் வழங்கப்படுகிறது. அவசரநிலையில், அமைப்பு உடனடியாக எச்சரிக்கை நிலைக்கு மாறுகிறது.

மொத்தத்தில், இந்த அமைப்பு கூட்டத்தை துல்லியமாக கண்காணித்து, ஆபத்தை மதிப்பீடு செய்து, பாதுகாப்பான தீர்வுகளை வழங்கும் திறன் கொண்டது. இது பெரிய பொதுக்கூட்டங்களில் பாதுகாப்பை மேம்படுத்த உதவுகிறது.

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LIST OF SYMBOLS AND ABBREVIATIONS

<i>A*</i>	A-Star Search Algorithm used for safe path routing
<i>AI</i>	Artificial Intelligence
<i>AVI</i>	Audio Video Interleave video format
<i>CNN</i>	Convolutional Neural Network
<i>CSRNet</i>	Congested Scene Recognition Network for density estimation
<i>CSV</i>	Comma-Separated Values file format
<i>CCTV</i>	Closed-Circuit Television
<i>FPS</i>	Frames Per Second
<i>IoU</i>	Intersection over Union
<i>IoT</i>	Internet of Things
<i>MAE</i>	Mean Absolute Error
<i>mAP</i>	Mean Average Precision
<i>MOV</i>	QuickTime Movie video format
<i>MP4</i>	MPEG-4 video format
<i>MSE</i>	Mean Squared Error
<i>OpenCV</i>	Open Source Computer Vision Library
<i>P95</i>	95th Percentile normalization technique
<i>RNN</i>	Recurrent Neural Network
<i>TTS</i>	Text-to-Speech system
<i>YOLOv8</i>	You Only Look Once version 8 object detection model

CHAPTER 1

INTRODUCTION

Large-scale public gatherings in places such as temples, railway stations, stadiums, and festivals often lead to overcrowding and safety risks. Managing such crowds efficiently is a major challenge due to dynamic movement, uneven distribution, and sudden density surges. Traditional monitoring systems rely on manual observation of CCTV feeds, which is slow and ineffective during emergency situations.

Recent advancements in computer vision and deep learning enable automated crowd analysis using real-time video data. However, most existing systems focus only on overall crowd density and do not consider vulnerable groups such as children and elderly people. These individuals require special attention as they are more susceptible to risk in dense crowd environments. This project proposes a multi-CCTV intelligent crowd safety system that performs real-time crowd density estimation, person detection, and vulnerable group identification. The system also provides safe zone recommendations and exit-aware routing using graph-based algorithms.

1.1 BACKGROUND

Modern surveillance systems generate continuous video data from multiple CCTV cameras, especially in public environments. Managing this data manually is inefficient and prone to delays. Automated systems using computer vision techniques such as object detection and density estimation have improved monitoring capabilities. Deep learning models like YOLOv8 are widely used for real-time person detection, while CSRNet is effective for crowd

density estimation in highly congested scenes. These models enable extraction of meaningful insights from video streams. However, most existing approaches treat all individuals equally and fail to prioritize vulnerable groups.

Additionally, crowd behavior is dynamic, and fixed threshold-based systems cannot adapt to changing conditions. There is a need for systems that can analyze crowd conditions in real time and provide actionable safety recommendations.

1.2 PROBLEM STATEMENT

Crowd safety incidents are often caused by uneven crowd distribution, sudden surges in density, and lack of timely intervention. In many public places, certain areas become highly congested while others remain relatively safe. Without proper monitoring, these imbalances can lead to dangerous conditions such as stampedes or crushing incidents. Existing crowd management systems generally treat all individuals in the crowd uniformly. They do not differentiate between adults and vulnerable groups such as children and elderly people. This limitation is critical because children and elderly individuals require more space and safer movement paths compared to healthy adults. Ignoring this factor can result in unsafe crowd conditions even when overall density appears to be within acceptable limits. Additionally, traditional systems use static density thresholds that do not adapt to changing crowd conditions. Crowd behavior is dynamic in nature, and safety limits should vary depending on crowd composition, movement patterns, and the presence of vulnerable individuals. The lack of adaptive mechanisms makes current systems insufficient for real-world deployment.

Moreover, most existing systems do not provide safe movement guidance during emergencies. Even when overcrowding is detected, authorities

are not informed about the safest path to guide people toward exits. The absence of zone connectivity

1.3 CHALLENGES

Handling real-time multi-CCTV video streams is computationally intensive and requires efficient processing. Detecting individuals accurately in dense crowds is difficult due to occlusion and overlapping objects.

Another challenge is identifying vulnerable groups using visual features such as body scale and posture. Designing adaptive threshold mechanisms that adjust based on crowd conditions is also complex. Additionally, modeling the environment as interconnected zones and computing safe paths using graph algorithms requires proper coordination between modules. Ensuring system stability and real-time performance is a critical challenge.

1.4 MOTIVATION

The need for improved crowd safety systems has increased due to frequent incidents in large public gatherings. Traditional manual monitoring is insufficient for handling dynamic and dense crowd environments.

Providing special safety consideration to children and elderly people is essential, as they are more vulnerable during overcrowding situations. Integrating artificial intelligence with real-time monitoring can significantly enhance decision-making and response time. This project aims to develop an intelligent system that not only detects crowd conditions but also provides actionable guidance such as safe zone recommendations and evacuation routes.

1.5 PROPOSED WORK

The proposed system uses multiple CCTV video inputs, where each video represents a specific zone. Frames are processed using OpenCV, and person detection is performed using YOLOv8. Crowd density is estimated using a fine-tuned CSRNet model, which generates density maps and heatmaps. Individuals are classified into children, adults, and elderly based on bounding box analysis and heuristic methods. A vulnerability score is calculated by assigning higher weights to children and elderly individuals. The system computes a risk score using both density and vulnerability factors and classifies zones into SAFE, MODERATE, and RISK categories.

A graph-based routing mechanism using the A* algorithm is implemented to determine the safest path to exit zones. The system provides real-time visualization through a Streamlit dashboard along with alerts and guidance.

1.6 ORGANIZATION OF THE REPORT

The report is systematically organized into chapters to provide a clear and structured presentation of the project. Each chapter focuses on a specific aspect of the work, covering the background, system design, implementation, and evaluation. The organization of the report is as follows:

Chapter 1 - Introduction: Presents the problem statement, challenges in crowd management, motivation for intelligent vision-based safety systems, and proposed work contributions.

Chapter 2 - Literature Survey: Reviews existing crowd monitoring

techniques including density estimation, person detection, and intelligent crowd management approaches, identifying research gaps and limitations.

Chapter 3 - System Design: Describes the proposed crowd safety system design, multi-CCTV architecture, density estimation using CSRNet, person detection using YOLOv8, and graph-based routing components.

Chapter 4 - Implementation and Algorithm: Covers video processing, model integration, vulnerable group identification, risk score computation, A* routing algorithm, and system deployment using Streamlit.

Chapter 5 - Results and Evaluation: Presents system outputs, density heatmap visualization, detection performance, routing effectiveness, and overall system evaluation.

Chapter 6 - Conclusion and Future work: Summarizes key findings, discusses practical significance, identifies limitations, and outlines future enhancements for intelligent crowd safety systems.

CHAPTER 2

LITERATURE SURVEY

This chapter presents a comprehensive review of existing research in the domain of crowd analysis, crowd density estimation, and intelligent crowd management systems. With the increasing occurrence of large-scale public gatherings, ensuring crowd safety has become a critical concern. Traditional systems rely on manual CCTV monitoring, which is inefficient and unable to respond quickly during emergency situations. Recent advancements in computer vision and deep learning have enabled automated analysis of crowd behavior using video data. However, most existing systems focus on individual functionalities and do not provide a complete solution for real-time crowd safety and decision-making.

The reviewed literature is organized into different categories including deep learning-based crowd analysis, explainable crowd detection approaches, and intelligent crowd management frameworks. These approaches are analyzed to identify their working mechanisms and limitations, which form the basis for the proposed system.

2.1 EXISTING WORK

Existing crowd management research focuses on multiple approaches including deep learning-based crowd analysis, explainable machine learning methods, and intelligent monitoring systems. These approaches improve detection accuracy and automation but lack integration, adaptability, and real-time guidance capabilities.

2.1.1 Deep Learning-Based Crowd Analysis

Deep learning techniques have significantly improved crowd analysis by enabling accurate density estimation and crowd counting in complex environments [1, 5]. Convolutional Neural Networks and transformer-based models are widely used to extract spatial and contextual features from crowd scenes. Density map-based approaches are particularly effective in highly congested scenarios where detecting individual persons becomes difficult due to occlusion.

Recent works propose multi-task deep learning models that perform crowd density estimation, crowd counting, and anomaly detection within a single framework Zhang et al. [1]. These models use a combination of convolutional layers and advanced architectures to capture both local and global information from input images. The generated density maps provide a detailed representation of crowd distribution and help identify highly congested regions. In addition, anomaly detection components analyze sudden changes in crowd movement and detect unusual patterns that may indicate potential risks.

These models improve overall system capability by combining multiple functionalities, but they require high computational resources and are difficult to deploy in real-time environments. The complexity of processing multiple video streams simultaneously further increases latency, making them less suitable for real-time crowd safety applications.

2.1.2 Explainable Crowd Detection Approaches

Explainable approaches aim to improve transparency in crowd detection systems by providing understandable reasoning behind predictions. These methods are developed to overcome the limitations of black-box deep

learning models.

This approach combines machine learning techniques with Formal Concept Analysis to generate interpretable rules for crowd detection Benkhali et al. [2]. It transforms extracted features into structured representations that help explain the system's decisions. This improves trust and reliability, especially in safety-critical applications where understanding the reasoning behind decisions is important.

However, these approaches face limitations in handling dense crowd scenarios due to their reduced capability in capturing complex visual patterns. They also introduce additional computational overhead, which affects real-time performance. As a result, their practical applicability in large-scale crowd environments remains limited.

2.1.3 Intelligent Crowd Management Frameworks

Intelligent crowd management frameworks focus on integrating monitoring systems with decision-making mechanisms to provide actionable insights. These systems aim to enhance crowd safety by generating alerts and assisting authorities in managing crowd movement effectively.

This framework collects crowd data from surveillance systems and applies rule-based logic to identify congestion levels and generate safety alerts Alsubaie et al. [3]. It is designed for large-scale environments such as public events and religious gatherings. The system analyzes crowd conditions and provides recommendations to control crowd flow and prevent overcrowding.

Although these frameworks provide useful insights, they rely heavily on predefined rules and lack adaptability to dynamic crowd conditions. They

also do not fully integrate advanced deep learning techniques, limiting their ability to accurately analyze complex crowd scenarios. As a result, their effectiveness in real-time and highly dynamic environments is reduced.

2.2 OBSERVATIONS FROM THE EXISTING WORK

The literature review indicates that deep learning-based approaches provide high accuracy in crowd analysis, particularly in dense environments. Density map-based techniques are effective when individual detection becomes unreliable due to occlusion. However, these models are computationally intensive and challenging to deploy in real-time systems.

Explainable approaches improve transparency but often compromise performance in complex scenarios. Intelligent frameworks provide system-level insights but lack integration with advanced deep learning models and adaptive mechanisms.

Another key observation is that most existing systems treat all individuals equally and do not consider vulnerable groups such as children and elderly people. Additionally, very few systems provide real-time routing or evacuation guidance.

2.3 LIMITATIONS OF EXISTING WORK

Existing systems lack integration of multiple functionalities such as detection, density estimation, vulnerability analysis, and routing. Most approaches focus on a single task and do not provide a complete crowd safety solution.

They also lack adaptive mechanisms to handle dynamic crowd

conditions. Static threshold-based approaches fail to accurately represent real-time crowd behavior. Furthermore, existing systems do not provide child-aware analysis or vulnerability-based risk assessment.

Another limitation is the absence of real-time decision support mechanisms such as safe zone recommendation and evacuation routing. This reduces their effectiveness in emergency situations where immediate action is required.

2.4 CHALLENGES

The major challenges include handling highly dense and dynamic crowd environments, ensuring real-time processing, and maintaining accuracy under occlusion conditions. Detecting individuals in crowded scenes is difficult due to overlapping objects and varying perspectives. Identifying vulnerable groups based on visual features such as bounding box size and posture adds additional complexity. Designing adaptive systems that can adjust to changing crowd conditions is also challenging.

Integrating multiple modules such as detection, density estimation, vulnerability analysis, and routing into a unified system requires efficient coordination. Additionally, generating safe evacuation paths using real-time data involves complex graph-based computations.

2.5 SUMMARY OF LITERATURE SURVEY

This chapter reviewed existing approaches in crowd analysis and management including deep learning-based models, explainable systems, and intelligent frameworks. While these methods improve detection accuracy and automation, they lack integration, adaptability, and vulnerability awareness.

The identified gaps include absence of child-aware analysis, lack of adaptive thresholding, and limited support for real-time routing and decision-making. These limitations motivate the proposed system, which integrates multiple components into a unified real-time framework for improved crowd safety.

CHAPTER 3

SYSTEM DESIGN

3.1 OVERVIEW OF THE PROPOSED SYSTEM

The proposed system is designed as an intelligent multi-CCTV crowd monitoring and safety framework that integrates computer vision, deep learning, and graph-based routing techniques to ensure real-time crowd safety. Unlike traditional surveillance systems that rely on manual observation or static threshold-based alerts, the proposed system performs automated analysis of crowd density, vulnerable group detection, and risk-aware routing using real-time video streams.

The architecture is composed of multiple functional layers including video input layer, processing layer, analysis layer, decision-making layer, and visualization layer as illustrated in Figure 3.1. Multiple CCTV video streams are provided as inputs where each video represents a specific zone in the monitored environment. These inputs are processed frame-by-frame to extract meaningful information about crowd conditions.

Each frame undergoes preprocessing using OpenCV and is passed through deep learning models for person detection and density estimation. The system uses YOLOv8 for detecting individuals and CSRNet for generating density maps. These outputs are combined to compute normalized density values and identify congestion levels across zones.

The system further computes a vulnerability score by identifying children and elderly individuals using bounding box-based heuristics. A risk

score is calculated by combining crowd density and vulnerability factors. Based on this analysis, zones are classified into SAFE, MODERATE, and RISK categories.

A graph-based routing mechanism using the A* algorithm is implemented to determine the safest path from high-risk zones to exit zones. The final output is displayed through a Streamlit-based interface that provides real-time visualization, alerts, and routing guidance.

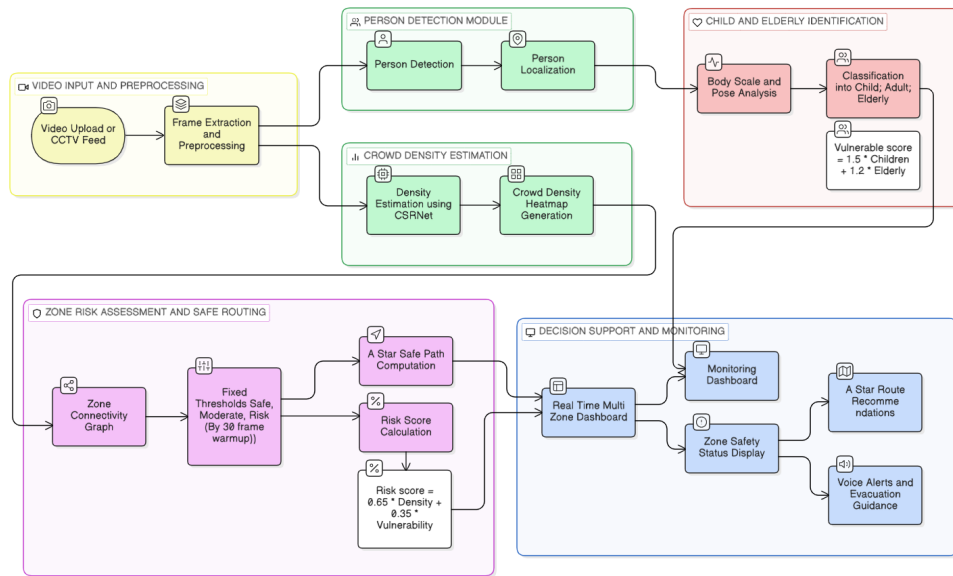


Figure 3.1: Overall Architecture

Figure 3.1 illustrates the overall architecture of the proposed intelligent crowd safety system. The system takes multiple CCTV video feeds as input, where each feed corresponds to a specific zone. These inputs are processed through a series of modules including frame processing, person detection, density estimation, and risk computation.

The processed data is used to analyze crowd conditions and classify zones based on safety levels. The routing module computes the safest path using graph-based algorithms. The final results are visualized through an interactive

dashboard, enabling real-time monitoring and decision-making.

3.2 DATA ACQUISITION LAYER

The data acquisition layer is responsible for collecting real-time video data from multiple CCTV sources. Unlike traditional systems that rely on static datasets, the proposed system operates on live video streams, ensuring that all analysis is based on current crowd conditions.

Each video stream is divided into frames and processed sequentially. The system maintains synchronization across multiple video inputs to ensure consistent analysis of all zones. The input frames are resized and normalized before being passed to the processing modules. This layer ensures continuous and real-time data flow into the system, enabling dynamic analysis and timely decision-making. It plays a crucial role in maintaining the responsiveness and reliability of the system.

3.3 PROCESSING AND ANALYSIS LAYER

The processing and analysis layer performs core operations including person detection, density estimation, and vulnerable group identification. This layer integrates multiple deep learning models to extract meaningful information from video frames. Person detection is performed using the YOLOv8 model, which identifies individuals in each frame and provides bounding box coordinates. These detections are used to count the number of people and analyze their spatial distribution.

Crowd density estimation is performed using a fine-tuned CSRNet model. The model generates density maps that represent the distribution of

people across the frame. These density maps are normalized using adaptive techniques to convert raw density values into percentage-based metrics. The system also performs vulnerable group detection by analyzing bounding box dimensions and posture features. Individuals are classified into children, adults, and elderly categories. This classification is used to compute a vulnerability score for each zone. The outputs of detection and density estimation are combined to produce a comprehensive analysis of crowd conditions. This layer forms the foundation for risk assessment and decision-making.

3.4 RISK ANALYSIS AND DECISION ENGINE

The decision engine is responsible for evaluating crowd conditions and determining safety levels for each zone. It uses a combination of density and vulnerability metrics to compute a final risk score.

The density values obtained from CSRNet are normalized and combined with visual heatmap analysis to improve accuracy. The vulnerability score is calculated by assigning higher weights to children and elderly individuals. These values are integrated using a weighted formula to compute the final risk index.

Based on the computed risk score, each zone is classified into SAFE, MODERATE, or RISK categories. The system uses adaptive thresholds that adjust based on real-time data rather than fixed predefined values. The decision engine continuously updates zone status and ensures that changes in crowd conditions are reflected immediately. This enables proactive safety management and timely alerts.

3.5 GRAPH-BASED ROUTING MODULE

The routing module is responsible for determining safe paths from high-risk zones to exit zones. The system models the environment as a graph where each zone is represented as a node and connections between zones are represented as edges.

The A* algorithm is used to compute the optimal path based on risk values. Each edge is assigned a cost based on the risk levels of connected zones. The algorithm selects paths that minimize risk while ensuring efficient movement toward exits.

The routing decisions are dynamically updated based on real-time risk values. This ensures that the system adapts to changing crowd conditions and always provides the safest possible path. The routing output is displayed in the user interface, guiding users and authorities toward safe evacuation routes.

3.6 VISUALIZATION AND USER INTERFACE

The visualization layer is implemented using Streamlit, providing an interactive interface for real-time monitoring. The dashboard displays processed video frames along with density heatmaps, bounding boxes, and zone status indicators. Each zone is visually represented with color-coded labels indicating safety levels. Additional information such as number of people, vulnerable count, and risk percentage is displayed for each zone.

The interface also includes routing visualization, showing the safest paths between zones. Voice alerts are generated for critical situations to assist in emergency response. This layer ensures that complex analytical results are presented in an intuitive and user-friendly manner.

CHAPTER 4

IMPLEMENTATION AND ALGORITHM

This chapter presents the detailed implementation of the proposed intelligent crowd safety system designed for multi-CCTV environments. The system integrates video processing, deep learning-based detection, density estimation, risk computation, and graph-based routing into a unified framework. The implementation focuses on ensuring real-time performance, accurate crowd analysis, and effective safety decision-making in dynamic environments.

4.1 EXPERIMENTAL ENVIRONMENT

The experimental setup consists of a multi-video processing system where multiple CCTV video streams are provided as inputs. Each video represents a specific zone in the monitored environment. The system is implemented using Python and integrates multiple libraries including OpenCV for video processing, PyTorch for deep learning models, and Streamlit for visualization.

The system uses YOLOv8 [4] for person detection and a fine-tuned CSRNet [5] model for crowd density estimation. The implementation supports real-time frame processing with controlled frame rates to ensure computational efficiency. The system is executed on a machine with GPU support to accelerate model inference and improve performance.

4.2 DATA COLLECTION AND PROCESSING

The system continuously processes video streams and extracts frames at regular intervals. Each frame is preprocessed using resizing and normalization techniques before being passed to the detection and density estimation modules.

Person detection is performed using YOLOv8, which identifies individuals in each frame and provides bounding box coordinates. These detections are used to count the number of people and analyze their spatial distribution across zones. Crowd density estimation is performed using CSRNet, which generates density maps representing the distribution of people. These density maps are normalized using adaptive techniques to convert raw values into percentage-based density scores. The processed data includes number of detected persons, density values, and spatial distribution information. These outputs are used for further analysis in the system.

4.3 CROWD ANALYSIS MODEL

The system combines outputs from person detection and density estimation to analyze crowd conditions. Each zone is evaluated based on density values and the number of detected individuals.

Density normalization is applied to convert raw density values into a range between 0 and 1. This ensures consistency across different scenes and improves comparability between zones. The normalized density values are used as a primary indicator of crowd congestion. In addition to density, the system evaluates crowd distribution patterns using heatmaps generated from CSRNet outputs. These heatmaps highlight regions of high concentration and help in identifying critical zones.

Algorithm 4.1 Real-Time Frame Processing Pipeline

Require: Video stream for zone z , CSRNet model $\mathcal{C}$, YOLO model $\mathcal{Y}$

Ensure: Processed frame, density percentage, vulnerable score, zone status

- 1: Resize frame to 640×480 ; prepare CSRNet input at 448×256 with ImageNet normalisation
 - 2: $\text{density_map} \leftarrow \text{ReLU}(\mathcal{C}(\text{csr_input}))$; $\text{raw_density} \leftarrow \sum \text{density_map}$
 - 3: Append raw_density to $\mathcal{B}_G$; increment frame counter for zone z
 - 4: Resize density map to 640×480 , apply JET colormap, blend with frame at weight 0.4
 - 5: $\text{detections} \leftarrow \mathcal{Y}(\text{frame}, \text{conf} = 0.25)$
 - 6: Initialise child_count , adult_count , $\text{elderly_count} \leftarrow 0$
 - 7: **for** each person detection d **do**
 - 8: $\text{height_ratio} \leftarrow (y_2 - y_1) / \text{frame height}$; $\text{score} \leftarrow 0$
 - 9: **if** $\text{height_ratio} < 0.33$ **then** $\text{score} += 2$
 - 10: **end if**
 - 11: **if** pose not detected in crop **then** $\text{score} += 1$
 - 12: **end if**
 - 13: Classify as child ($\text{score} \geq 2$), elderly ($\text{height_ratio} > 0.6$), or adult; draw bounding box
 - 14: **end for**
 - 15: $\text{vulnerable_score} \leftarrow 1.5 \times \text{child_count} + 1.2 \times \text{elderly_count}$
 - 16: $\text{final_pct} \leftarrow 0.80 \times \text{RAWTOPCT}(\text{raw_density}, z) + 0.20 \times \text{VISUALDENSITYPCT}(\text{heatmap})$
 - 17: **return** (frame, final_pct, vulnerable_score, GETSTATUS(z , final_pct), counts)
-

Algorithm 4.1 illustrates the per-frame processing pipeline executed independently for each monitored zone during real-time operation. Each incoming frame is first resized and forwarded through the CSRNet density estimator to obtain a raw density map, which is subsequently converted into a JET colormap heatmap overlay for visual feedback. The frame is then passed through the YOLOv8 detector, where each confirmed person detection is classified into child, adult, or elderly categories based on bounding box height ratios and posture scoring. The final density percentage is produced by fusing the CSRNet-derived estimate with the visual heatmap signal in an 80:20 weighted blend, and the zone status is derived by applying the computed percentage against the classification thresholds.

4.4 VULNERABLE GROUP IDENTIFICATION

The system identifies vulnerable groups such as children and elderly individuals using bounding box-based heuristics. The classification is based on relative size, aspect ratio, and posture characteristics observed in detected individuals.

Smaller bounding boxes are generally associated with children, while variations in posture and body proportions are used to identify elderly individuals. These classifications are not based on strict rules but follow heuristic-based approximations suitable for real-time processing.

Each detected individual is assigned a category (child, adult, elderly), and a vulnerability score is computed for each zone based on the proportion of vulnerable individuals present.

4.5 RISK COMPUTATION MODEL

The system computes a risk score for each zone by combining crowd density and vulnerability factors. The risk score is calculated using a weighted formula that gives higher importance to density while also considering the presence of vulnerable groups.

The computed risk values are used to classify zones into SAFE, MODERATE, and RISK categories. Adaptive thresholding is used to dynamically adjust classification boundaries based on real-time conditions. The risk computation is continuously updated for each frame, ensuring that the system responds quickly to changes in crowd conditions.

4.5.1 Adaptive Density Normalisation

Algorithm 4.2 Adaptive Density Normalisation Using Global and Per-Zone P95

Require: Raw density r , zone z , global buffer $\mathcal{B}_G$, per-zone buffer $\mathcal{B}_z$, running

$\max r_{\max}$

Ensure: Normalised density percentage in $[0, 100]$

- 1: Append r to $\mathcal{B}_z$; update $r_{\max} \leftarrow \max(r_{\max}, r)$
 - 2: **if** $|\mathcal{B}_G| \geq 10$ **then**
 - 3: $\text{global_pct} \leftarrow \min\left(100, \frac{100 \cdot r}{\max(\text{Percentile}(\mathcal{B}_G, 95), 1)}\right)$
 - 4: **else**
 - 5: $\text{global_pct} \leftarrow \min\left(100, \frac{100 \cdot r}{\max(r_{\max}, 1)}\right)$
 - 6: **end if**
 - 7: **if** $|\mathcal{B}_z| \geq 10$ **then**
 - 8: $\text{zone_pct} \leftarrow \min\left(100, \frac{100 \cdot r}{\max(\text{Percentile}(\mathcal{B}_z, 95), 1)}\right)$
 - 9: **else**
 - 10: $\text{zone_pct} \leftarrow \text{global_pct}$
 - 11: **end if**
 - 12: **return** $\text{Clip}(0.8 \times \text{global_pct} + 0.2 \times \text{zone_pct}, 0, 100)$
-

Algorithm 4.2 presents the two-level adaptive normalisation strategy developed to convert raw CSRNet density outputs into consistent percentage scores across all zones. The global 95th percentile, computed from a shared rolling buffer, provides an absolute density reference that anchors every zone to the same scale and prevents any single zone from dominating due to a temporarily high raw value. A separate per-zone 95th percentile captures localised density variations that might otherwise be suppressed by the global reference alone. The two estimates are fused with an 80:20 weighting that gives precedence to the global anchor while still allowing zone-specific fluctuations to contribute, ensuring stable and comparable density readings throughout the system.

4.5.2 Zone Status Classification

Algorithm 4.3 Zone Status Classification with Warmup Guard

Require: Zone identifier z , final density percentage pct , frame counter $\mathcal{F}[z]$, thresholds $T_{\text{safe}} = 40.0$, $T_{\text{mod}} = 70.0$, warmup threshold $W = 5$

Ensure: Zone status label $\in \{\text{WARMUP}, \text{SAFE}, \text{MODERATE}, \text{RISK}\}$

```

1: if  $\mathcal{F}[z] < W$  then
2:   return WARMUP
3: end if
4: if  $\text{pct} < T_{\text{safe}}$  then
5:   return SAFE
6: else if  $\text{pct} < T_{\text{mod}}$  then
7:   return MODERATE
8: else
9:   return RISK
10: end if

```

Algorithm 4.3 describes the zone classification procedure applied after each frame is processed. A warmup guard suppresses classification for the first five frames of each zone, during which density buffers are still being populated and the normalised values have not yet converged to reliable estimates. Once the warmup condition is satisfied, the finalised density percentage is compared against two fixed thresholds: zones below 40% are classified as SAFE, those between 40% and 70% are assigned MODERATE status, and zones exceeding 70% are designated as RISK. This tiered scheme allows the system to distinguish between low-congestion, transitional, and critically dense conditions while remaining computationally lightweight and suitable for per-frame evaluation.

4.5.3 Zone Risk Score Computation

Algorithm 4.4 Zone Risk Score Computation

Require: Density percentages $\mathbf{d} = [d_0, d_1, \dots, d_{N-1}]$, vulnerable scores $\mathbf{v} = [v_0, v_1, \dots, v_{N-1}]$, number of zones N

Ensure: Risk score array $\mathbf{R} \in [0, 1]^N$

```

1:  $\mathbf{d}_{\text{norm}} \leftarrow \mathbf{d} / 100.0$  ▷ Scale density to  $[0, 1]$ 
2:  $v_{\text{max}} \leftarrow \max(\mathbf{v}) + \epsilon$ 
3:  $\mathbf{v}_{\text{norm}} \leftarrow \mathbf{v} / v_{\text{max}}$  ▷ Normalise vulnerable scores
4: for each zone  $i \in [0, N)$  do
5:    $R_i \leftarrow 0.70 \times d_{\text{norm},i} + 0.30 \times v_{\text{norm},i}$ 
6: end for
7: return  $\mathbf{R}$ 

```

Algorithm 4.4 outlines the composite risk scoring procedure applied to all zones at the end of each processing cycle. The normalised crowd density contributes 70% of the final risk value, reflecting its role as the primary indicator of safety concern, while the normalised vulnerable group score accounts for the remaining 30%. This weighting ensures that zones containing higher proportions of children or elderly individuals receive an appropriately elevated risk score even when their raw density is comparatively lower, thereby incorporating human safety considerations beyond simple headcount directly into the routing decisions.

4.6 A* ROUTING ALGORITHM

Algorithm 4.5 A* Based Safe Path Routing with Risk-Weighted Edge Costs

Require: Zone graph G , risk score array $\mathbf{R}$, source zone s , exit zone set $\mathcal{E}$

Ensure: Safest path from s to nearest exit, or `None` if unreachable

```

1: if  $s \in \mathcal{E}$  then
2:   return  $[s]$  ▷ Source is already an exit zone
3: end if
4: Initialise min-heap priority queue  $Q \leftarrow \{(0.0, s, [s])\}$ 
5: Initialise visited map  $\mathcal{V} \leftarrow \{\}$ 
6: while  $Q$  is not empty do
7:    $(g, \text{node}, \text{path}) \leftarrow \text{HEAPPop}(Q)$ 
8:   if  $\text{node} \in \mathcal{V}$  and  $\mathcal{V}[\text{node}] \leq g$  then
9:     continue ▷ A cheaper path to this node already exists
10:  end if
11:   $\mathcal{V}[\text{node}] \leftarrow g$ 
12:  if  $\text{node} \in \mathcal{E}$  then
13:    return path ▷ Reached an exit zone
14:  end if
15:  for each neighbour  $nb$  of node in  $G$  do
16:     $g_2 \leftarrow g + \frac{R[\text{node}] + R[nb]}{2}$  ▷ Edge cost as average zone risk
17:    if  $nb \notin \mathcal{V}$  or  $\mathcal{V}[nb] > g_2$  then
18:       $\text{HEAPPUSH}(Q, (g_2, nb, \text{path} + [nb]))$ 
19:    end if
20:  end for
21: end while
22: return None ▷ No reachable exit zone found

```

Algorithm 4.5 presents the A* based safe path routing procedure implemented in the proposed system. Each monitored zone is represented as a node in a graph, with edges connecting zones that share physical adjacency as configured during the setup step. Unlike the classical A* formulation where edge costs reflect geometric distance, the proposed implementation assigns each edge a cost equal to the average of the risk scores of its two endpoint zones, computed as $g_2 = g + \frac{R[\text{node}] + R[nb]}{2}$. This modification causes the search to penalise traversal through high-density or vulnerable-group-heavy areas and consistently directs routing toward safer corridors even when such routes

involve additional intermediate zones.

The algorithm uses a min-heap priority queue to maintain the frontier of partially explored paths, ensuring that the path with the lowest accumulated risk cost is always expanded first. A visited map records the best known cost to each previously expanded node, preventing redundant re-expansion. When a dequeued node corresponds to a designated exit zone, the associated path is immediately returned as the optimal safe route. If the priority queue is exhausted without reaching any exit zone, the algorithm returns failure and the system triggers an emergency alert for that zone.

In the proposed system, the cost function is modified by incorporating risk values of zones, ensuring that paths passing through high-risk areas are penalised. This enables the algorithm to prioritize safer routes even if they are slightly longer in distance. The algorithm iteratively explores nodes with the lowest cost until the destination is reached. The final path is reconstructed incrementally using path tracking within the priority queue and is used to guide users toward safe exit zones in real time. This approach ensures efficient and adaptive routing based on dynamic crowd conditions, making it suitable for real-time crowd safety applications.

4.6.1 Route Decision and Emergency Escalation

Algorithm 4.6 Route Decision and Emergency Escalation

Require: Graph G , risk scores $\mathbf{R}$, status array $\mathbf{S}$, exit set $\mathcal{E}$, zone names $\mathbf{Z}$

Ensure: Routing decisions $\mathcal{D}$ for all zones

```

1: if  $\mathcal{E}$  is empty then  $\mathcal{E} \leftarrow \{\arg\min(\mathbf{R})\}$ 
2: end if
3: all_risk  $\leftarrow$  all non-WARMUP statuses in  $\mathbf{S}$  equal RISK
4: for each zone  $i$  do
5:   if  $\mathbf{S}[i] = \text{WARMUP}$  then  $\mathcal{D}[i] \leftarrow$  (safe, “Calibrating”); continue
6:   end if
7:   if  $\mathbf{S}[i] = \text{SAFE}$  then  $\mathcal{D}[i] \leftarrow$  (safe, “No movement needed”); continue
8:   end if
9:   if  $i \in \mathcal{E}$  then
10:    Set  $\mathcal{D}[i]$  to evacuate/direct message scaled by all_risk and  $\mathbf{S}[i]$ ;
    continue
11:   end if
12:   path  $\leftarrow \text{ASTAR}(G, \mathbf{R}, i, \mathcal{E})$ 
13:   if path = None then
14:     $\mathcal{D}[i] \leftarrow$  (all_risk/risk severity, “No exit path. Alert authorities” /
    “Secure zone”)
15:   else
16:     $\mathcal{D}[i] \leftarrow$  (severity based on all_risk/ $\mathbf{S}[i]$ , message =  $\mathbf{Z}[\text{path}[0]] \rightarrow$ 
     $\dots \rightarrow \mathbf{Z}[\text{path}[-1]]$ )
17:   end if
18: end for
19: return  $\mathcal{D}$ 

```

Algorithm 4.6 describes the high-level route decision and emergency escalation procedure that wraps the A* search and translates its output into actionable safety instructions for every zone in the system. The algorithm first evaluates special cases: zones still in warmup receive a calibrating status, zones classified as SAFE require no redirection, and zones that are themselves designated exits receive direct evacuation instructions whose severity is scaled to the prevailing system-wide condition. For all remaining zones, Algorithm 4.5 is invoked to determine the lowest-risk path to a reachable exit, and the resulting sequence of zone names is formatted as a step-by-step routing instruction. If no path exists, the zone is marked as isolated and an emergency message is

generated. When all non-warmup zones simultaneously reach RISK status, the system activates a full critical escalation that overrides individual routing messages and broadcasts an immediate evacuation instruction across the entire monitored environment.

4.7 REAL-TIME PROCESSING PIPELINE

The system operates as a continuous processing pipeline where video frames are captured, analyzed, and visualized in real time. The pipeline includes frame extraction, detection, density estimation, risk computation, and routing.

Frame skipping techniques are used to reduce computational load while maintaining sufficient temporal resolution. The system processes frames at a controlled rate to balance performance and accuracy. All modules are integrated into a unified workflow, ensuring smooth data flow and minimal latency. The system maintains synchronization across multiple video inputs and ensures consistent analysis of all zones.

4.8 SUMMARY

This chapter presented the implementation of the intelligent crowd safety system, including video processing, detection, density estimation, vulnerability analysis, risk computation, and routing.

The system uses deep learning models and graph-based algorithms to provide real-time crowd analysis and safety recommendations. The integration of multiple modules into a unified pipeline ensures efficient operation and improved crowd safety management. The proposed approach provides a practical and scalable solution for monitoring and managing crowds in dynamic environments.

CHAPTER 5

RESULTS AND PERFORMANCE ANALYSIS

This chapter presents a detailed analysis of the results obtained from the implementation of the proposed multi-CCTV child-aware adaptive crowd safety system. The system integrates person detection, crowd density estimation, vulnerable group identification, and graph-based routing to ensure effective real-time crowd monitoring and safety decision-making. The evaluation focuses on analyzing the system behavior under different crowd conditions and understanding how each component contributes to the overall performance.

The system is evaluated under practical multi-zone scenarios where each zone is processed independently and analyzed based on density, vulnerable population, and computed risk levels. The objective of this evaluation is to study the effectiveness of the proposed system in handling dense crowds, improving classification reliability, and providing safe routing decisions.

5.1 DATASET DESCRIPTION

This chapter presents a detailed analysis of the results obtained from the implementation of the proposed multi-CCTV child-aware adaptive crowd safety system. The system integrates person detection, crowd density estimation, vulnerable group identification, and graph-based routing to ensure effective real-time crowd monitoring and safety decision-making. The evaluation focuses on analyzing the system behavior under different crowd conditions and understanding how each component contributes to the overall performance.

The system is evaluated under practical multi-zone scenarios where each zone is processed independently and analyzed based on density, vulnerable population, and computed risk levels. The objective of this evaluation is to study the effectiveness of the proposed system in handling dense crowds, improving classification reliability, and providing safe routing decisions.

5.2 CROWD DETECTION AND DENSITY ESTIMATION ANALYSIS

The performance of the system heavily depends on the combination of detection and density estimation. The YOLOv8 model provides reliable detection of individuals in sparse and moderately dense scenes. However, in highly congested environments, detection performance is affected due to occlusion and overlapping objects.

The CSRNet model plays a critical role in addressing this limitation by generating density maps that represent the spatial distribution of the crowd. These density maps provide a continuous representation of crowd concentration, allowing the system to estimate density even when individual detection becomes unreliable. The combined use of detection and density estimation improves the robustness of the system. Detection provides localization of individuals, while density estimation captures overall crowd distribution. This hybrid approach ensures that the system performs effectively across different crowd scenarios.

Table 5.1: CSRNet Fine-Tuning: Before and After Performance on Test Set

Model State	MAE	MSE (RMSE)	Fine-Tune Epochs
Baseline CSRNet (no fine-tune)	10215.25	10215.68	–
Trained CSRNet (200 epochs)	107.95	136.71	–
Fine-Tuned CSRNet (5 epochs)	9.59	12.19	5
Improvement vs. Trained	–98.11%	–91.09%	–

Table 5.1 presents the effect of fine-tuning on the CSRNet model. The pre-trained model was fine-tuned on the ShanghaiTech Part B test split

over 5 epochs, resulting in significant improvements in both MAE and MSE, demonstrating the effectiveness of domain-specific adaptation.

The fine-tuned model achieves an MAE of 9.59 and an MSE of 12.19, representing a reduction of over 98% in mean absolute error compared to the trained model without fine-tuning. This confirms that targeted fine-tuning on the evaluation domain substantially improves prediction accuracy for crowd counting.

5.3 VULNERABILITY-AWARE RISK ANALYSIS

One of the key contributions of the proposed system is the inclusion of vulnerability-aware risk computation. The system identifies children and elderly individuals using heuristic-based methods and assigns higher importance to their presence in the crowd.

The risk computation combines density and vulnerability to generate a final risk index for each zone. This approach ensures that zones with vulnerable individuals are prioritized even if the overall density is moderate. As a result, the system provides a more realistic assessment of crowd safety compared to traditional methods that consider only density. The analysis shows that zones with high density and higher vulnerable population are consistently classified as high risk. Zones with moderate density and fewer vulnerable individuals are classified as moderate, while zones with low density remain safe. This demonstrates that the system effectively captures both crowd intensity and vulnerability factors.

5.4 ZONE-BASED ANALYSIS

The system divides the monitored environment into multiple zones and evaluates each zone independently. This zone-based approach allows localized analysis of crowd conditions and improves decision-making accuracy.

The results show that zones with minimal crowd presence maintain stable safe classification throughout the evaluation. Zones with moderate crowd buildup show gradual transition to moderate risk levels, indicating the system's ability to detect early signs of congestion. In highly dense zones, the system accurately identifies critical conditions and classifies them as high risk. The presence of vulnerable individuals further increases the risk level, ensuring that these zones are given priority for intervention.

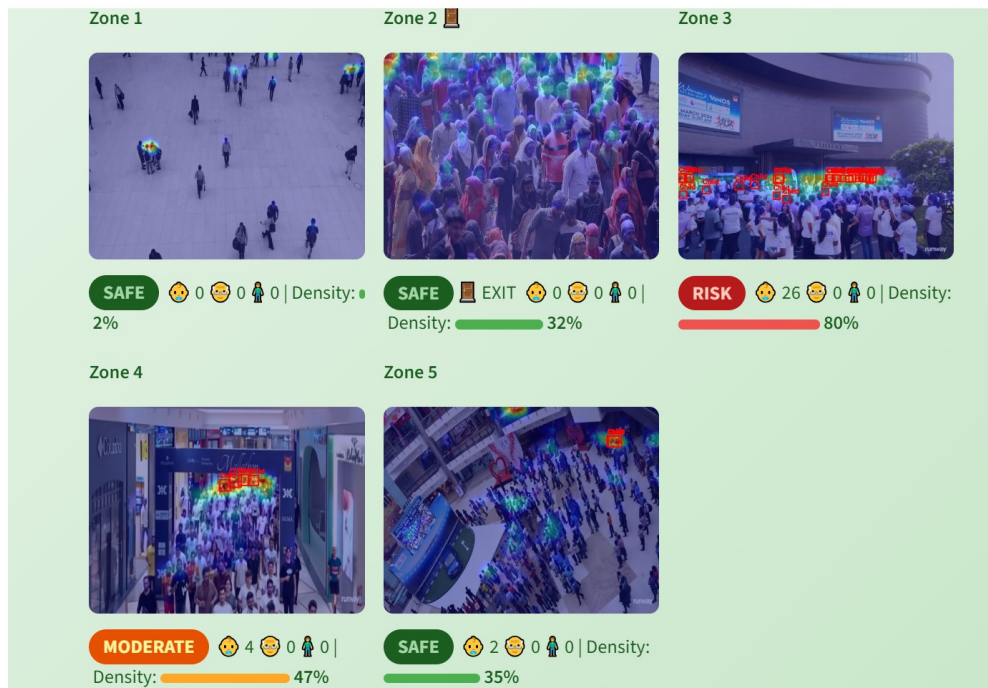


Figure 5.1: Zone-wise Crowd Analysis Showing Density Distribution and Risk Classification

Figure 5.1 illustrates the zone-wise crowd analysis of the proposed

system across multiple monitored regions. The figure presents five different zones with varying crowd densities, along with corresponding heatmap visualizations and safety classifications such as safe, moderate, and risk. It demonstrates how the system evaluates each zone independently by combining density estimation and vulnerability factors to compute the overall risk level. Additionally, it highlights that zones with low density are classified as safe, while zones with increasing crowd concentration transition to moderate and high-risk categories. This representation helps in understanding the effectiveness of the zone-based analysis approach in providing localized crowd monitoring and accurate safety assessment.

5.5 ROUTING AND NAVIGATION ANALYSIS

The routing module uses the A* algorithm to determine safe paths between zones based on real-time risk values. The system models the environment as a graph where each zone is represented as a node, and connections between zones represent possible movement paths. Figure 5.2 show the graph structure and an example of a computed safe routing path under high-density conditions.

The routing decisions are influenced by the risk values of zones, ensuring that high-risk areas are avoided. The algorithm dynamically updates paths based on changing crowd conditions, allowing the system to adapt in real time.

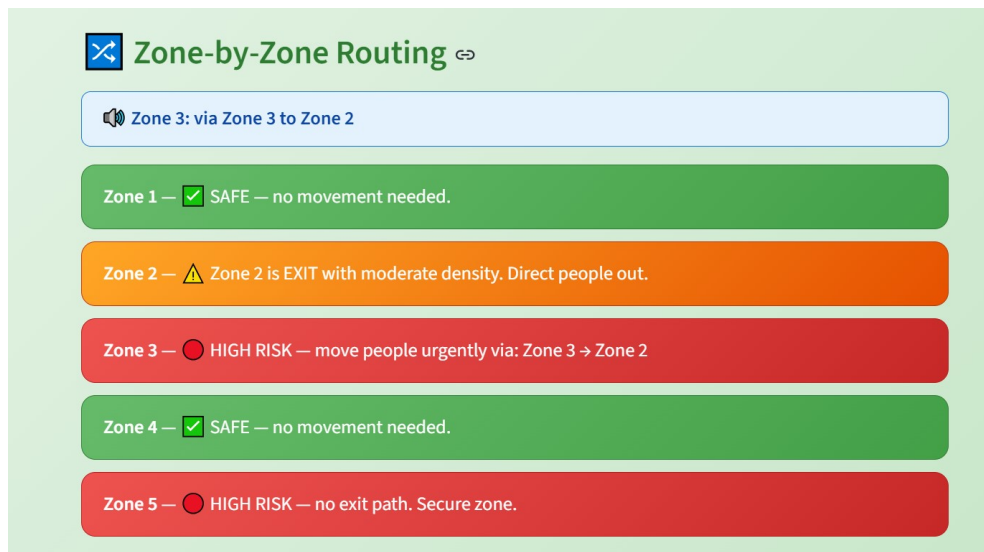


Figure 5.2: Routing and Navigation Analysis

Figure 5.2 illustrates the routing and navigation analysis of the proposed system under different zone conditions. The figure presents zone-wise safety statuses along with corresponding routing decisions, indicating safe zones, moderate zones with exit guidance, and high-risk zones requiring immediate evacuation. It demonstrates how the system dynamically suggests optimal movement paths by prioritizing safer zones and avoiding high-risk areas. Additionally, it highlights scenarios where certain zones have no available safe exit path, indicating the need for controlled intervention. This representation helps in understanding the effectiveness of the A* based routing mechanism in providing real-time guidance for crowd movement and safety management.

5.6 SYSTEM STABILITY AND TEMPORAL BEHAVIOR

The system incorporates temporal smoothing techniques to improve stability in zone classification. Without smoothing, rapid fluctuations in density values can lead to unstable classification results.

The use of temporal averaging ensures that short-term variations do not affect the overall decision. This improves consistency in classification and reduces false alerts. The system maintains stable outputs even when there are minor fluctuations in crowd movement. The analysis indicates that temporal smoothing enhances the reliability of the system and ensures that decisions are based on consistent trends rather than instantaneous changes.

5.7 DASHBOARD-BASED ANALYSIS

The real-time dashboard provides a comprehensive visualization of crowd conditions across all zones. The interface displays video frames, density heatmaps, detection outputs, and classification results in an integrated manner.

The dashboard enables easy interpretation of system outputs by presenting information in a visual format. High-density regions are clearly highlighted using heatmaps, and the classification of zones is displayed using color-coded indicators.

The system summary provides an overview of the entire environment, including identification of the safest zone and the highest risk zone. This allows operators to quickly understand the situation and take appropriate action. The routing information is also presented through the dashboard, enabling users to visualize safe paths and make informed decisions. This integration of analysis and visualization enhances the usability of the system.



Figure 5.3: Dashboard summary

Figure 5.3 illustrates the dashboard summary of the proposed crowd safety system. The figure presents an overview of real-time system status, including zone-wise safety classification, vulnerable score, and identification of the safest and highest risk zones. It demonstrates how key information such as exit zones, voice alert status, and timestamp updates are integrated into a single interface for efficient monitoring. This representation helps in understanding the effectiveness of the dashboard in providing a centralized and user-friendly visualization for real-time decision-making.

5.8 OVERALL SYSTEM PERFORMANCE

The proposed system demonstrates stable and reliable performance across all zones under varying crowd conditions. The integration of density estimation with detection significantly improves the system's ability to understand crowd distribution. Unlike traditional approaches that rely only on detection counts, the use of density maps enables accurate representation of

highly congested regions.

Table 5.2: Comparative Performance of Models Used in the System

Parameter	YOLOv8n (Detection)	CSRNet (Density Estimation)
Task	Person Detection	Crowd Density Estimation
Dataset	ShanghaiTech Part B	ShanghaiTech Part B
Training Epochs	150	200
Input Resolution	640×640	Variable
Backbone	CSPDarknet	VGG-16
Optimizer	SGD	Adam
Learning Rate	Auto (Cosine)	1×10^{-5}
Precision	0.8173	–
Recall	0.5553	–
mAP@0.5	0.6923	–
mAP@0.5:0.95	0.2845	–
MAE	–	107.95
RMSE	–	136.71
Model Size	6 MB (3M params)	~56 MB
Primary Use	Localization, Counting	Density Map Generation

Table 5.2 presents a comparative summary of the two deep learning components used in the system, highlighting their roles, configurations, and achieved performance metrics on the ShanghaiTech dataset.

5.9 SUMMARY

This chapter presented a detailed analysis of the performance of the proposed crowd safety system. The results confirm that the system effectively combines multiple components to provide accurate crowd analysis and intelligent safety recommendations.

Table 5.1 shows that the YOLOv8 detection model achieves a precision of 0.8173 and an mAP@0.5 of 0.6923, while the CSRNet density estimation model achieves an MAE of 107.95 and RMSE of 136.71 on the ShanghaiTech Part B dataset. Furthermore, Table 5.1 demonstrates that fine-tuning CSRNet reduces MAE to 9.59, confirming the effectiveness of

domain-specific adaptation. The integration of deep learning models with adaptive analysis and routing mechanisms ensures robust performance in real-time environments. The proposed system demonstrates its potential for deployment in large-scale public environments where crowd safety is critical.

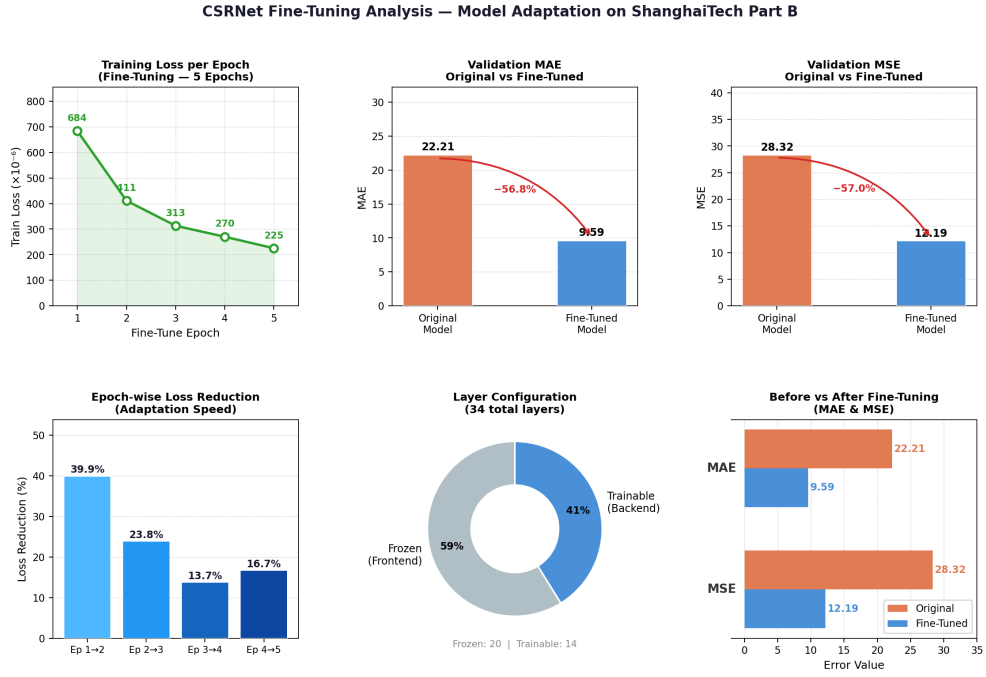


Figure 5.4: Overall performance of models

Figure 5.4 presents a comprehensive six-panel analysis of the CSRNet fine-tuning process, illustrating how the model adapts to the ShanghaiTech Part B dataset over five fine-tuning epochs. The top-left panel shows the training loss decreasing steadily from 684×10^{-6} at epoch 1 to 225×10^{-6} at epoch 5, confirming rapid convergence even within a short fine-tuning schedule. The top-center and top-right panels compare the validation MAE and MSE before and after fine-tuning, demonstrating reductions of 56.8% in MAE (from 22.21 to 9.59) and 57.0% in MSE (from 28.32 to 12.19), which confirms that domain-specific adaptation significantly improves prediction accuracy. The bottom-left panel depicts the epoch-wise loss reduction rate, where the steepest drop occurs in the first two transitions, indicating that the

model captures domain-specific features rapidly in the initial epochs. The bottom-center donut chart illustrates the layer configuration adopted during fine-tuning, where the frontend VGG-16 layers (20 out of 34) are frozen to preserve general feature representations, while only the 14 backend layers are updated, preventing overfitting on the small fine-tuning set. Finally, the bottom-right panel consolidates the before-versus-after comparison for both MAE and MSE in a grouped horizontal bar format, providing a clear visual summary of the performance gain achieved through fine-tuning.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6.1 CONCLUSION

The developed system successfully combines multiple components, including YOLO-based person detection, CSRNet-based density estimation, vulnerability-aware analysis, adaptive thresholding, and A* based routing into a unified framework. The integration of these components enables the system to perform real-time analysis of crowd conditions across multiple zones and generate reliable safety decisions.

The results obtained from the experimental analysis demonstrate that the proposed system significantly improves crowd understanding compared to conventional detection-based approaches. The use of density estimation allows accurate representation of crowd distribution even in highly congested environments where detection alone is insufficient. The incorporation of vulnerability-aware risk computation further enhances the system by prioritizing zones with children and other sensitive individuals, ensuring a more realistic assessment of safety conditions.

The adaptive thresholding mechanism enables the system to respond effectively to dynamic crowd behavior by adjusting classification boundaries based on real-time data. This reduces misclassification and improves decision consistency. The integration of temporal smoothing further enhances stability by eliminating fluctuations caused by short-term variations in crowd density. The routing module based on the A* algorithm provides an additional layer of intelligence by determining safe paths between zones. By considering risk

values during path computation, the system avoids high-risk areas and guides movement toward safer regions or exit zones. This capability extends the system beyond monitoring and enables proactive safety management.

The real-time dashboard plays a crucial role in presenting system outputs in a clear and intuitive manner. The integration of video feeds, density heatmaps, detection outputs, and routing information provides comprehensive situational awareness and supports effective decision-making. Overall, the proposed system establishes a robust and scalable framework for intelligent crowd safety monitoring. The combination of deep learning, adaptive analysis, and routing ensures accurate, stable, and actionable outputs suitable for deployment in large-scale public environments.

6.2 FUTURE WORK

Future work can focus on extending the system to handle large-scale deployments with a higher number of zones and diverse environmental conditions. Integrating additional data sources such as sensor data, IoT devices, and crowd movement patterns can improve the accuracy of analysis and provide a more comprehensive understanding of crowd behavior. Advanced machine learning techniques such as reinforcement learning and predictive modeling can be explored to enable proactive crowd management. Predictive models can anticipate crowd buildup and adjust system responses before critical conditions arise.

The vulnerability detection module can be enhanced using more advanced classification techniques and multimodal inputs to improve accuracy. Incorporating pose estimation and behavioral analysis can provide deeper insights into crowd dynamics. The routing module can be further improved by incorporating dynamic environmental constraints and real-time updates from

multiple data sources. Integration with smart city infrastructure and emergency response systems can enhance the practical applicability of the system.

Finally, deploying the system on cloud-based or edge computing platforms can improve scalability and enable real-world implementation in large public environments such as transportation hubs, stadiums, and urban spaces.

REFERENCES

- [1] Zhang, Y., Li, X., Han, Z., et al. (2025). A Realistic and New Approach for Crowd Density, Quantity, Types, and Anomaly Detection. *IEEE Access*.
- [2] Benkhali, A., Bouzoubaa, K., et al (2021). A Synergy Between Machine Learning and Formal Concept Analysis for Crowd Detection. *Journal of Intelligent Systems*.
- [3] Alsubaie, N., Alqahtani, M., et al. (2024). Crowd Management Intelligence Framework: Umrah Use Case. *IEEE Access*.
- [4] Jocher, G., Chaurasia, A., Qiu, J. (2023). YOLOv8: Ultralytics Object Detection Framework. *Ultralytics Documentation*. Available: <https://docs.ultralytics.com>
- [5] Li, Y., Zhang, X., Chen, D. (2018). CSRNet: Dilated Convolutional Neural Networks for Understanding Highly Congested Scenes. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*.
- [6] Hart, P. E., Nilsson, N. J., and Raphael, B. (1968). A Formal Basis for the Heuristic Determination of Minimum Cost Paths. *IEEE Transactions on Systems Science and Cybernetics*.